



中国科学技术大学

数理方程 UNIVERSITY OF SCIENCE AND TECHNOLOGY OF CHINA

第六周作业

Hefei, Anhui, 230026 The People's Republic of China

18. 解: (1)
$$\begin{cases} u_{tt} + \frac{1}{r} u_r + u_{zz} = 0 & (0 < r < a, 0 < z < l) \\ u(a, z) = 0 \\ u(r, 0) = 0, u(r, l) = T_0 \end{cases}$$

设 $u(r, z) = R(r)Z(z)$

$$\Rightarrow \frac{R''(r)}{R(r)} + \frac{1}{r} \frac{R'(r)}{R(r)} + \frac{Z''(z)}{Z(z)} = 0$$

设 $\frac{R''(r)}{R(r)} + \frac{1}{r} \frac{R'(r)}{R(r)} = -\lambda$

$R(r)$ 满足 $\begin{cases} r^2 R''(r) + r R'(r) + \lambda r^2 R(r) = 0 \\ R(a) = 0, R(0) \text{ 有界} \end{cases} \Rightarrow \begin{cases} \lambda_n = \omega_n^2 & n \geq 1 \text{ 其中 } \omega_n \text{ 为 } J_0(\omega_n a) = 0 \text{ 正根} \\ R_n(r) = J_0(\omega_n r) \end{cases}$

λ_n 代入 $\begin{cases} Z''(z) - \lambda Z(z) = 0 \\ Z(0) = 0 \end{cases}$ 得 $Z_n(z) = C_n \sinh(\omega_n z)$

故 $u(r, z) = \sum_{n=1}^{\infty} C_n \sinh(\omega_n z) J_0(\omega_n r)$

代入 $u(r, l) = T_0$ 得 $C_n \sinh(\omega_n l) = \frac{\int_0^a J_0(\omega_n r) \cdot T_0 r dr}{\int_0^a J_0^2(\omega_n r) r dr} = \frac{2T_0}{\omega_n a J_1(\omega_n a)}$

故 $u(r, z) = \sum_{n=1}^{\infty} \frac{2T_0}{\omega_n a J_1(\omega_n a)} \frac{\sinh(\omega_n z)}{\sinh(\omega_n l)} J_0(\omega_n r)$

(2) $\begin{cases} u_{tt} + 2h u_t = a^2 (u_{rr} + \frac{1}{r} u_r) & (h < c) \\ u(0, t) \text{ 有限 } u_r(l, t) = 0 \\ u(r, 0) = \varphi(r), u_t(r, 0) = 0 \end{cases}$

设 $u(r, t) = R(r)T(t)$

$$\Rightarrow \frac{T''(t)}{T(t)} + 2h \frac{T'(t)}{T(t)} = a^2 \left(\frac{R''(r)}{R(r)} + \frac{1}{r} \frac{R'(r)}{R(r)} \right)$$

设 $\frac{R''(r)}{R(r)} + \frac{1}{r} \frac{R'(r)}{R(r)} = -\lambda$

$R(r)$ 满足 $\begin{cases} r^2 R''(r) + r R'(r) + \lambda r^2 R(r) = 0 \\ R'(l) = 0, R(0) \text{ 有界} \end{cases} \Rightarrow \begin{cases} \lambda_0 = 0, R_0(r) \text{ 为常数} \\ \lambda_n = \omega_n^2, R_n(r) = J_0(\omega_n r) & (n \geq 1) \end{cases}$

ω_n 为 $J_1(\omega_n l) = 0$ 正根

λ_n 代入 $\begin{cases} T''(t) + 2h T'(t) + \lambda a^2 T(t) = 0 \\ T'(0) = 0 \end{cases}$ 得 $T_0(t) = C_0, T_n(t) = e^{-ht} (C_n \cos q_n t + D_n \sin q_n t)$

其中 $q_n = \sqrt{\omega_n^2 a^2 - h^2}, D_n = \frac{h}{q_n} C_n$

故 $u(r, t) = C_0 + \sum_{n=1}^{\infty} e^{-ht} (C_n \cos q_n t + D_n \sin q_n t) J_0(\omega_n r)$

代入 $u(r, 0) = \varphi(r)$ 得 $C_0 + \sum_{n=1}^{\infty} C_n J_0(\omega_n r) = \varphi(r) \Rightarrow \begin{cases} C_0 = \frac{2}{l^2} \int_0^l r \varphi(r) dr \\ C_n = \frac{2}{l^2} \frac{\int_0^l J_0(\omega_n r) \varphi(r) r dr}{J_1^2(\omega_n l)} & n \geq 1 \\ D_n = \frac{h}{q_n} C_n \end{cases}$





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20 解: $P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2-1)^n$; $w(x,t) = (1-2xt+t^2)^{-\frac{1}{2}} = \sum_{n=0}^{+\infty} P_n(x) t^n$

令 $x=0$ 有 $(1+t^2)^{-\frac{1}{2}} = \sum_{n=0}^{+\infty} P_n(0) \cdot t^n = \sum_{m=0}^{+\infty} \binom{-\frac{1}{2}}{m} t^{2m} \Rightarrow \begin{cases} P_{2m+1}(0) = 0 \\ P_{2m}(0) = \binom{-\frac{1}{2}}{m} = \frac{(-\frac{1}{2}) \cdots (-\frac{2m-1}{2})}{m!} \end{cases}$

化简后得 $P_n(0) = \begin{cases} 0, & n=2m+1 \\ \frac{(-1)^m (2m)!}{2^{2m} (m!)^2}, & n=2m \end{cases} \quad m \geq 0$

$\frac{\partial w}{\partial x} = \frac{t}{(1-2xt+t^2)^{3/2}} = \sum_{n=0}^{+\infty} P'_n(x) t^n$ 令 $x=0$ 有 $\sum_{n=0}^{+\infty} P'_n(0) t^n = \sum_{m=0}^{+\infty} \binom{\frac{1}{2}}{m} t^{2m+1}$

求得 $P'_n(0) = \begin{cases} 0, & n=2m \\ \frac{(-1)^m (2m+1)!!}{(2m)!!}, & n=2m+1 \end{cases} \quad m \geq 0$

22. 解: x^m 按勒让德多项式展开, 最高次数为 m 次, 即: $x^m = \sum_{k=0}^m a_k P_k(x)$, $a_k = \frac{2k+1}{2} \int_{-1}^1 x^m P_k(x) dx$
当 $m < n$ 时, $a_n = 0$, $\int_{-1}^1 x^m P_n(x) dx = 0$;

当 $m \geq n$ 时, $\int_{-1}^1 x^m P_n(x) dx = \frac{1}{2^n n!} \int_{-1}^1 x^m \frac{d^n}{dx^n} (x^2-1)^n dx \xrightarrow{\text{分部积分 } n \text{ 次}} \frac{(-1)^n}{2^n n!} \int_{-1}^1 \frac{d^n (x^m)}{dx^n} (x^2-1)^n dx$

$\Rightarrow \int_{-1}^1 x^m P_n(x) dx = \frac{(-1)^n}{2^n n!} \cdot \frac{m!}{(m-n)!} \int_{-1}^1 x^{m-n} (x^2-1)^n dx$ Beta 积分
 $= \frac{(-1)^n}{2^n n!} \frac{m!}{(m-n)!} (-1)^n B(k+\frac{1}{2}, n+1)$ (其中 $m-n=2k, k \in \mathbb{Z}$)
 $= \frac{2 \cdot m!}{(m-n)!! (m+n+1)!!}$

故 $\int_{-1}^1 x^m P_n(x) dx = \begin{cases} 0 & m < n \text{ 或 } m-n \text{ 为奇数} \\ \frac{2 \cdot m!}{(m-n)!! (m+n+1)!!} & m-n \text{ 为偶数且 } m \geq n \end{cases}$

(2) $x P_m(x) = \frac{m+1}{2m+1} P_{m+1}(x) + \frac{m}{2m+1} P_{m-1}(x)$ ($m \geq 0$, 规定 $P_{-1}(x) = 0$)

故积分非零 $\Leftrightarrow m-n = \pm 1$

若 $m = n+1$, $\int_{-1}^1 x P_m(x) P_n(x) dx = \int_{-1}^1 \frac{m}{2m+1} P_n^2(x) dx = \frac{2(n+1)}{(2n+1)(2n+3)}$

若 $m = n-1$ 类似地, $\int_{-1}^1 x P_m(x) P_n(x) dx = \frac{2n}{4n^2-1}$

$\int_{-1}^1 x^m P_n(x) dx = \begin{cases} 0, & m-n \neq \pm 1 \\ \frac{2(n+1)}{(2n+1)(2n+3)}, & m-n=1 \\ \frac{2n}{4n^2-1}, & m-n=-1 \end{cases}$





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24 解: (1) $f(x) = x^3 = \frac{2}{5} P_3(x) + \frac{3}{5} P_1(x)$

$(x^3 = \sum_{k=1}^3 a_k P_k(x) + 22(1))$

(2) $f(x) = x^4 = \frac{8}{35} P_4(x) + \frac{4}{7} P_2(x) + \frac{1}{5} P_0(x)$

$(x^4 = \sum_{k=1}^4 a_k P_k(x) + 22(1))$

(3) 设 $f(x) = |x| = \sum_{n=0}^{+\infty} a_n P_n(x)$ $f(x)$ 为偶函数 $\Rightarrow a_{2k+1} = 0 \quad (k \geq 0)$

$a_{2k} = \frac{4k+1}{2} \int_{-1}^1 |x| P_{2k}(x) dx = (4k+1) \int_0^1 x P_{2k}(x) dx = \frac{4k+1}{2k+2} \int_0^1 P_{2k-1}(x) dx \quad (k \geq 1)$

将 $P_{2k-1}(x) = \frac{1}{4k-1} (P_{2k}'(x) - P_{2k-2}'(x))$ 代入得 $a_{2k} = \frac{4k+1}{(2k+2)(4k-1)} [P_{2k-2}(0) - P_{2k}(0)]$

20题中结论代入 $\Rightarrow a_{2k} = (-1)^{k+1} \frac{4k+1}{2^{2k}(k-1)!} \frac{(2k-2)!}{(k+1)!} \quad k \geq 1; \quad a_0 = \frac{1}{2}$

$f(x) = \frac{1}{2} + \sum_{k=1}^{+\infty} (-1)^{k+1} \frac{(4k+1)(2k-2)!}{2^{2k}(k-1)!(k+1)!} P_{2k}(x)$

28 解: (1) $u(r, \theta): \begin{cases} \Delta u = 0 & 0 < r < a, 0 \leq \theta \leq \pi \\ u(a, \theta) = u_0 & 0 \leq \theta < \frac{\pi}{2} \\ u(a, \theta) = -u_0 & \frac{\pi}{2} < \theta \leq \pi \\ u(r, \frac{\pi}{2}) = 0 \end{cases}$

设 $u(r, \theta) = R(r) \Theta(\theta)$

代入有 $\frac{1}{R} \frac{d}{dr} (r^2 \frac{dR}{dr}) = -\frac{1}{\Theta \sin \theta} \frac{d}{d\theta} (\sin \theta \frac{d\Theta}{d\theta}) = \lambda$

$\Rightarrow \lambda_n = n(n+1) \quad R_n(r) = A_n r^n + \frac{B_n}{r^{n+1}}, \quad \Theta_n(\theta) = P_n(\cos \theta)$

球心处温度有限 $\Rightarrow B_n = 0$, 可设 $u(r, \theta) = \sum_{n=0}^{+\infty} C_n r^n P_n(\cos \theta)$

$r=a$ 处展开: $u(a, \theta) = \sum_{n=0}^{+\infty} C_n a^n P_n(\cos \theta) = \begin{cases} u_0 & 0 \leq \theta < \frac{\pi}{2} \\ 0 & \theta = \frac{\pi}{2} \\ -u_0 & \frac{\pi}{2} < \theta \leq \pi \end{cases}$

$\Rightarrow C_{2k+1} = \frac{u_0(4k+5)}{a^{2k+1}} \frac{(-1)^k (2k)!}{2^{2k+1} k! (k+1)!}$

$u(r, \theta) = u_0 \sum_{k=0}^{+\infty} \frac{4k+5}{2^{2k+1}} \frac{(2k)!}{k! (k+1)!} \left(\frac{r}{a}\right)^{2k+1} P_{2k+1}(\cos \theta)$

(2) $u(r, \theta): \begin{cases} \Delta u = 0 & 0 \leq r \leq a, 0 \leq \theta \leq \pi \\ u(a, \theta) = u_0 \end{cases}$

(底面绝热即 $P_n'(0) = 0$, n 为偶数时成立, 故只需偶延拓)

设 $u(r, \theta) = \sum_{n=0}^{+\infty} C_n r^n P_n(\cos \theta)$

$r=a$ 处展开 $\Rightarrow C_0 = u_0, C_n = 0 \quad (n \geq 1)$

故 $u(r, \theta) = u_0$

